Citrus Fruit Classification: Two-Approach Computer Vision Report

28 October 2026

Summary

On isolated-fruit features the SVM and the Random Forest tie for the best test macro-F1 (0.989, 98 of 99 images), but a lightweight CNN trained the same way is the model that stays accurate when the background changes (0.944, against 0.829 for the Random Forest and 0.169 for the SVM).

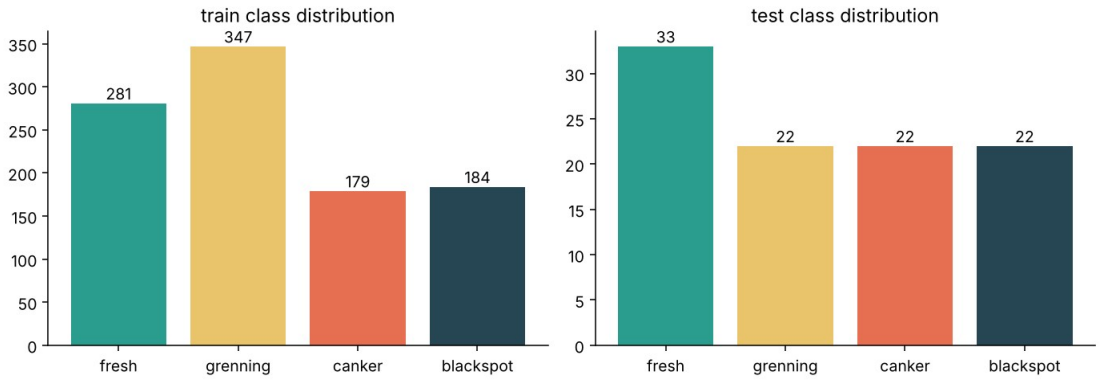
Dataset note. The data supplied ("Orange Dataset.zip", 1,090 images) is a four-class orange problem: fresh, grenning (citrus greening), canker and blackspot. It is not avocado ripening, so the same pipeline was run on this data. The method carries over unchanged: only the class names differ.

Two approaches were compared on one leak-safe split: **A** hand-crafted colour, texture and lesion features with SVM and Random Forest, and **B** a 211k-parameter CNN trained from scratch (3 seeds). Both were tested on the images as-is and on the fruit isolated from its background, to measure how much each model relies on background shortcuts.

1. Understanding of the dataset

The data has three problems that shape every modelling choice: moderate class imbalance, a strong source and background shortcut, and near-duplicate images that leak between train and test. No file is corrupt and no two files are byte-identical.

Class	Train	Test	File type	Typical background
fresh	281	33	PNG, 150+ different sizes	white or grey studio, leaves visible
grenning	347	22	JPG, 800×800	black
canker	179	22	JPG, 800×800	white or grey
blackspot	184	22	JPG, 800×800	white or grey



Class distribution in train and test

Class imbalance. In train the largest class (grenning, 347) is 1.9× the smallest (canker, 179). The test set is balanced differently (fresh 33%, the rest 22% each), so train and test class shares differ. I used class-weighted losses and report macro-F1 and per-class recall, not accuracy alone.

Outliers.

- Size: all 776 JPGs are exactly 800×800; the 314 fresh PNGs span about 150 to 585 px wide, and 312 of them are aspect-ratio outliers.
- Sharpness: 125 images are blur outliers by Laplacian variance, 123 of them fresh.
- Format: 32 images are RGBA with transparency, handled by compositing on white.

Leakage. 8 test images have a near-duplicate in train (perceptual hash distance ≤ 4), and 2 of those pairs carry different labels. Another 45 train images sit in near-duplicate groups, because many photos are rotated copies of one fruit. I dropped the 7 train images that match a test image and kept near-duplicate groups together when splitting train from validation.

Biases.

- File type alone identifies “fresh” (every fresh image is a PNG, no other class has one).
- Background identifies “grenning”: border brightness is about 9 of 255, against 117 to 148 for the others.
- Mean image brightness is 158 for fresh and 47 for grenning, so brightness separates them without looking at the fruit.
- Fresh images look like catalogue photos (smooth peel, leaves, no lesions); the three disease classes come from a different camera setup.
- Canker and blackspot share one photo setup, so they are the only pair where the model has to read the fruit itself.

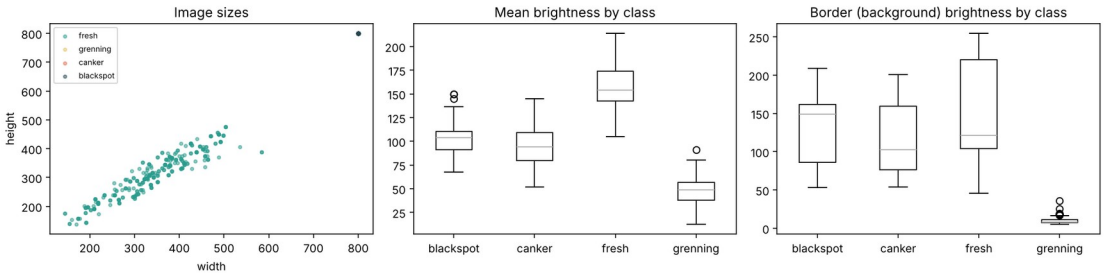


Image size, brightness and background brightness by class



Random training samples per class

2. Pipeline and preprocessing

Every model was trained and tested on the same leak-safe split, in two input variants, so the comparison isolates the effect of the model and of the background shortcut.

- **Split.** The official test set (99 images) is untouched. From train I removed the 7 images that duplicate a test image, then made a stratified split that keeps near-duplicate groups together: 819 images to fit, 165 for validation. Hyperparameters and the best CNN epoch were chosen on validation only.
- **Input S1, "as-is".** The whole image resized to 128×128; transparent PNGs are composited on white.
- **Input S2, "isolated".** The fruit is segmented (saturated pixels, largest connected blob, holes filled), cropped to a square and placed on a neutral grey background, then resized

to 128×128. This removes background colour, file-source borders and rotation padding. Segmentation fell back to the full image on 5 of 1,090 images.

- **Why two inputs.** If a model scores well on S1 but collapses on S2, it learned the background. That is the check I use to decide which result to trust.



Left of each pair: as-is input. Right: isolated fruit on neutral grey.

3. Model selection

I chose one classical and one deep approach because the dataset is small (about 1,000 images) and the classes differ in ways both can exploit: colour (green, yellow or orange peel) and lesions (spot count, size and texture).

Approach A: hand-crafted features + SVM / Random Forest.

- Features (94 per image): HSV colour histograms, colour moments in HSV and Lab, uniform LBP texture at two scales, dark-blob statistics at three thresholds (a lesion proxy), and green and orange pixel fractions.
- SVM (RBF kernel) suits small datasets with many correlated features; C and gamma were picked on validation. Random Forest needs no feature scaling, handles mixed feature types and gives feature importances.
- Both use balanced class weights. Fast (about 27 ms per image end to end on 2 CPU cores), explainable, and needs no GPU.

Approach B: lightweight CNN from scratch.

- Four blocks of two 3×3 convolutions with batch norm, ReLU and max pooling (16, 32, 64, 96 channels), global average pooling and dropout 0.3: 211k parameters, a 0.86 MB file.
- It learns lesion shape and texture directly instead of relying on my feature design, and it is small enough for a phone or a sorting line.
- Augmentation: flips, 90° rotations, random crop-and-resize, mild brightness and contrast jitter. Hue is left alone because colour is part of the signal.

- Class-weighted cross-entropy with label smoothing 0.05, AdamW with a one-cycle schedule, 25 epochs, best epoch chosen on validation macro-F1, 3 seeds.
- Trained from scratch because downloading pretrained weights was blocked in this environment. With about 1,000 images that is a real limit, which is why the network is kept small; MobileNetV2 transfer learning is the first improvement listed in section 7.

4. Evaluation metrics

I ranked models by macro-F1, then checked per-class recall, the confusion matrix and a background-swap test, because accuracy alone hides imbalance and the 99-image test set is small.

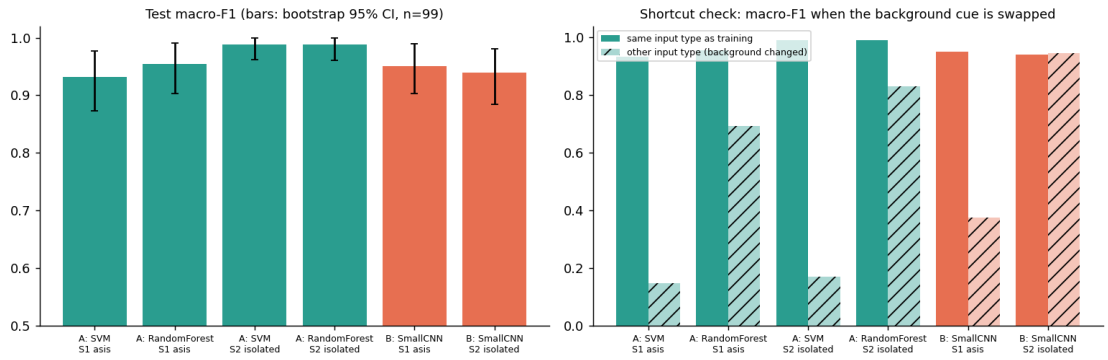
- **Accuracy.** Reported for reference only. With uneven classes and 99 test images, one mistake moves it by about one point.
- **Macro-F1 (the main metric).** The unweighted mean of per-class F1, so the smaller classes (canker, blackspot) count as much as the larger ones. It also balances precision and recall in one number.
- **Per-class recall, lowest class.** A diseased fruit passed as healthy is the costly error in a sorting line, so I read the weakest class's recall, not just the average.
- **Confusion matrix.** Shows which classes are confused. Here it shows that nearly all errors are canker versus blackspot.
- **Bootstrap 95% confidence interval on macro-F1** (1,000 resamples of the test set). Needed because with n=99 two models can differ by a few points and still be indistinguishable.
- **Background-swap macro-F1.** Each model is also scored on the other input type (as-is vs isolated). A large drop means it learned the background, not the fruit.
- **Cost.** Latency per image and model size, since the use case is a production line or a phone.

How the choice was made. A model wins only if it has the higher macro-F1 and a lowest-class recall that is not worse, and if it does not collapse under the background swap. Where confidence intervals overlap, I say so instead of declaring a winner on test score alone.

5. Results

The SVM and the Random Forest on isolated-fruit features tie for the best test score (macro-F1 0.989, one error in 99 images), and the CNN trained on isolated fruit is the one that holds up when the background changes. All six configurations score between 0.93 and 0.99, and most confidence intervals overlap.

Model	Input	Accuracy	Macro-F1 (95% CI)	Lowest-class recall	Macro-F1 after background swap
A: SVM	as-is	0.939	0.932 (0.873–0.977)	0.818	0.148
A: Random Forest	as-is	0.960	0.955 (0.903–0.990)	0.909	0.693
A: SVM	**isolated**	**0.990**	**0.989 (0.961–1.000)**	**0.955**	0.169
A: Random Forest	**isolated**	**0.990**	**0.989 (0.960–1.000)**	**0.955**	0.829
B: CNN, mean of 3 seeds	as-is	0.956	0.951 ± 0.014	0.849	0.374
B: CNN, mean of 3 seeds	isolated	0.946	0.939 ± 0.011	0.788	**0.944**



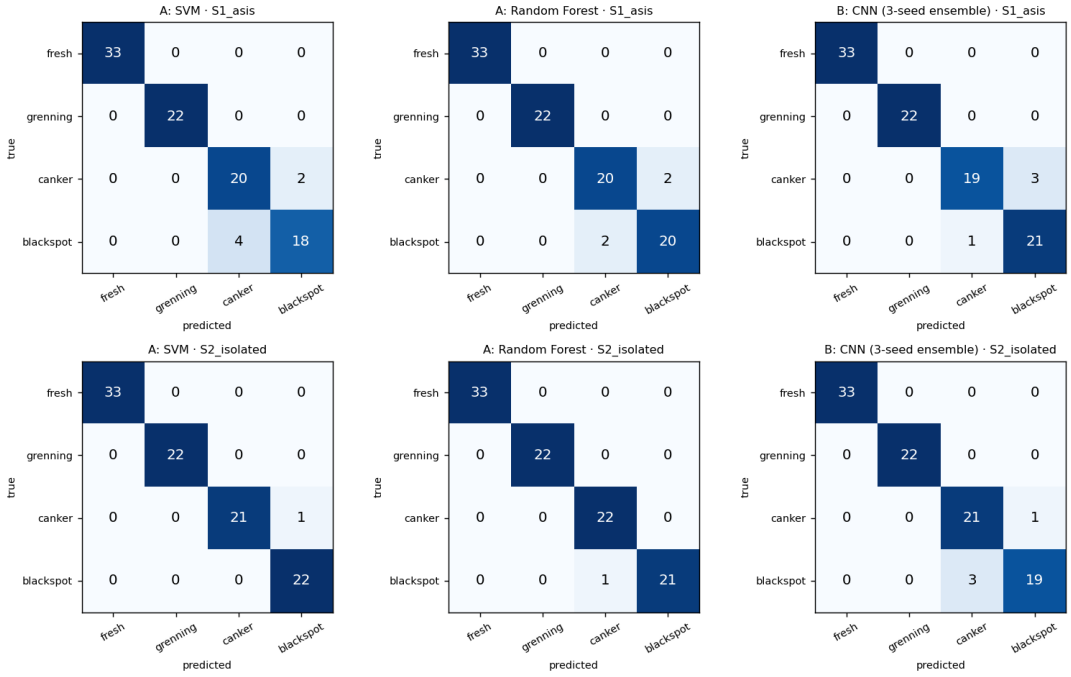
Test macro-F1 with bootstrap intervals, and the background-swap check

What the numbers say.

- **Winner on score.** The SVM and the Random Forest on isolated fruit tie, with one error each in 99 images, and lead on every test metric. The CNN on the same input makes about 3 to 4 more mistakes (its 3-seed ensemble makes 4 errors in total), but the

intervals overlap, so this is suggestive, not proven. Averaging the 3 CNN seeds gives 0.954 macro-F1, still below the two classical models.

- **Where the errors are.** The CNN ensemble's 4 mistakes on isolated fruit are all canker versus blackspot (3 blackspot called canker, 1 canker called blackspot), the one pair that shares a photo setup. That pair is the real task; fresh and grenning are partly solved by the shortcuts described in section 1. The confusion matrices below show the same pattern for every other model: fresh and grenning are classified perfectly in all six, and every remaining mistake is canker versus blackspot.
- **Shortcut check.** The CNN trained on as-is images falls from 0.951 to 0.374 macro-F1 when the background is removed (0.24 to 0.49 across seeds): it learned the background. The CNN trained on isolated fruit scores 0.944 on as-is images, so it learned the fruit. The SVM collapses in both directions (0.148 and 0.169) and the Random Forest drops to 0.693 and 0.829. The classical numbers are partly inflated by a feature shift, since colour histograms are computed on fruit pixels only for isolated input, so the CNN comparison is the cleaner evidence.
- **Seed noise.** CNN macro-F1 on isolated fruit ranged from 0.931 to 0.954 across three seeds, about half the gap to the classical models. The same code on a different library version also moved the SVM from 0.911 to 0.989, so differences of a few points should not be over-read.
- **Cost.** In my environment CNN inference took 4.7 to 6.2 ms per image on 2 CPU cores, and the classical pipeline about 27 ms end to end because of feature extraction. Segmentation time was not measured separately.



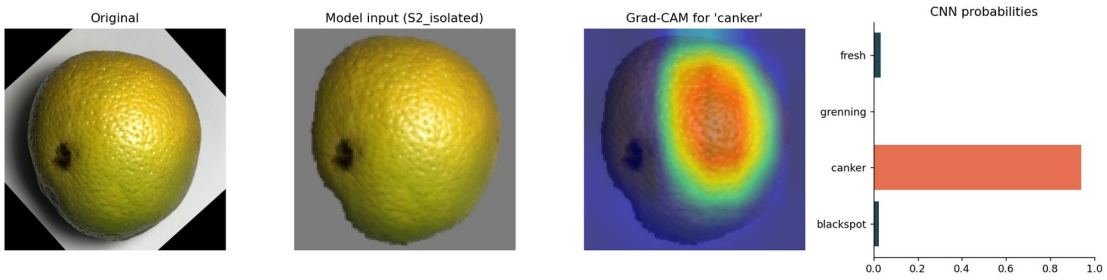
Confusion matrices: rows are as-is input, then isolated input; columns are SVM, Random Forest, CNN ensemble

6. Example run

On a held-out canker image all three models are right; on a held-out blackspot image the CNN is wrong with 85% confidence while both classical models are right.

```
python predict.py "test/canker/c (1).jpg" --scn S2_isolated --viz
example_canker.png

A:RandomForest        -> canker   conf 0.95
A:SVM                 -> canker   conf 0.72
B:CNN(ensemble of 3)  -> canker   conf 0.94
```

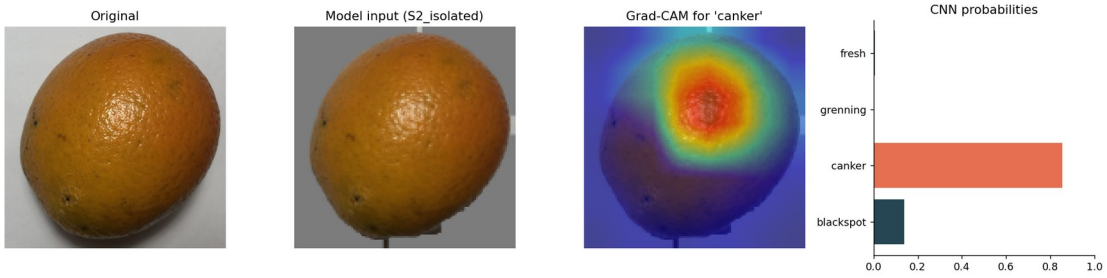


True class canker: original, model input, Grad-CAM, CNN probabilities

The Grad-CAM map shows the CNN looking at the textured centre-right of the peel. It does not point at the dark lesion on the left edge, so for this image the decision rests on peel texture and colour more than on the visible lesion.

```
python predict.py "test/blackspot/b (3).jpg" --scn S2_isolated --viz
example_error.png

A:RandomForest        -> blackspot conf 0.66
A:SVM                 -> blackspot conf 0.68
B:CNN(ensemble of 3)  -> canker    conf 0.85
```



True class blackspot, misclassified as canker by the CNN

This fruit has only a few tiny dark spots, and the CNN attends to the glossy centre instead of the spots. Both classical models, which count dark blobs explicitly, get it right. The confident wrong answer matters for production (section 7).

7. The four mandatory questions

1. How will the model perform in production?

Expect clearly lower accuracy than the 0.94 to 0.99 measured here, and treat that figure as an upper bound. The test images come from the same photo setups as the training images, and fresh fruit is a different source from the three disease classes.

- A model that sees raw images drops from 0.95 to 0.37 macro-F1 when only the background changes, so production needs the fruit-isolation step and a single fruit on a plain background.
- Speed is not the problem: in my environment the CNN needed about 5 to 6 ms per image on 2 CPU cores and is 0.86 MB; the classical pipeline needs about 27 ms.
- Healthy fruit photographed in a packing house will not look like the catalogue-style fresh images used here, so fresh versus disease errors that this test cannot show are likely.
- The CNN gave a wrong answer with 0.85 confidence, so confidence alone should not decide. Route low-confidence and canker-versus-blackspot cases to a person, and monitor the class mix and confidence over time.

2. What are the drawbacks of the model?

- The dataset contains shortcuts (file type, background, brightness) that no preprocessing fully removes, and the fresh and grenning scores are partly inflated by them.
- The test set has 99 images, so confidence intervals are about ±5 points and the models cannot be ranked with certainty.
- Perceptual hashing does not reliably match heavily rotated copies, so near-duplicate leakage may be understated and scores slightly optimistic.
- The CNN still confuses canker and blackspot (4 to 5 of 99 test images), and small lesions are lost when 800×800 images are shrunk to 128×128.
- The CNN is trained from scratch, varies between seeds (0.931 to 0.954 macro-F1 on isolated fruit), and is overconfident when wrong.
- Fruit isolation uses hand-set colour thresholds: it fell back to the full image on 5 images, keeps leaves, and would struggle with pale fruit or strong glare.
- Only four classes: a different disease or a non-orange input will be forced into one of them.

3. What would I improve with more time?

- Fine-tune a pretrained MobileNetV2 or EfficientNet-B0 (blocked here), then quantize it for edge use.
- Collect healthy fruit under the same camera and background as the disease classes, plus more canker and blackspot images. This removes the main shortcut at its source.
- Use repeated cross-validation grouped by fruit to get tighter intervals than a single 99-image test set allows.
- Replace threshold segmentation with GrabCut or a small segmentation network, and train on higher-resolution crops or lesion patches.

- Calibrate confidences (temperature scaling), add an “unknown” option, and ensemble the CNN with the classical models, which got the example error right.
- Have an expert review the canker and blackspot labels: 2 near-duplicate pairs across train and test carry conflicting labels.

4. Where did I spend the most time?

On understanding the data, not on training. Profiling the images, finding the file-type and background shortcuts, detecting duplicates and building the isolated-fruit comparison took most of the effort, because that analysis decided which results could be trusted. Training was cheap: feature extraction takes about 20 seconds, and the six CNN runs took about 1.5 hours on a laptop CPU, mostly unattended. The rest went on environment limits: the supplied data turned out to be oranges rather than avocados, and pretrained weights could not be downloaded.

Code and how to run

The scripts run in this order; paths (`ROOT` for the unzipped data, `OUT` for results) are set at the top of each file, and the split uses a fixed seed.

1. `eda.py`: dataset profiling, duplicates, outliers, figures.
2. `prep.py`: builds the as-is and isolated 128×128 arrays.
3. `classical.py`: features, SVM and Random Forest, background-swap test.
4. `cnn.py --scn S1_asis|S2_isolated --seed N --epochs 25`: trains the CNN.
5. `aggregate.py`: comparison table, confusion matrices, charts.
6. `predict.py <image> --scn S2_isolated --viz out.png`: the example run.

Packages: torch, torchvision, scikit-learn, scikit-image, opencv-python-headless, imagehash, matplotlib, pandas, joblib.